

Upstream Controller
(LQR, MPC, PPO)

u_{prop}

Robust HOCBF-QP
Safety Filter

u^* (safe)

Supercritical
Boiler-Turbine

x_{k+1}

$\epsilon(x)$ Margin
(σ -chain propagation)

$\sigma_{\text{GP}}(x)$

GP Residual Model
 $\mu_{\text{GP}}(x), \sigma_{\text{GP}}(x)$

Process Constraints

Pressure: 13–24 MPa
Enthalpy: 2670–2830 kJ/kg
Power: ± 50 MW

Real-Time Latency

QP solve: ~ 25 ms (GPU)

$\sim 10\times$ faster than SLSQP-based NMPC (254 ms)

Model Mismatch $\rightarrow$ Robust HOCBF-QP $\rightarrow$ No Observed CBF Violations

6 static perturbation scenarios $\cdot$ 5 seeds $\cdot$ 50 episodes $\cdot$ Wilson upper bound 1.46%